Amazon Subnet — Important Interview Questions & Answers

1. What is a Subnet?
A subnet is a smaller network created inside a VPC.
It helps us divide the VPC network into separate network sections.
Simple ga gurthupettuko:
Subnet ante VPC lopala create chese smaller network.
Eng: A subnet is a smaller network created inside a VPC.

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2. What is a Public Subnet?
A public subnet is a subnet whose route table has a route to an Internet Gateway.
Resources in this subnet can communicate with the internet when the required settings are configured.
Simple ga gurthupettuko:
Public Subnet ante Internet Gateway ki route unna subnet.
Eng: A public subnet is a subnet that has a route to the Internet Gateway.

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3. What is a Private Subnet?
A private subnet does not have a direct route to an Internet Gateway.
Resources in this subnet can use a NAT Gateway for outgoing internet access.
Simple ga gurthupettuko:
Private Subnet ante direct Internet Gateway route leni subnet.
Eng: A private subnet does not have a direct route to the Internet Gateway.

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4. What is the difference between Public and Private Subnet?
A public subnet has a route to an Internet Gateway, while a private subnet does not have a direct route to it.
Private subnet resources can use a NAT Gateway when they need internet access.
Simple ga gurthupettuko:
Public Subnet ki IGW route untundi, Private Subnet ki direct IGW route undadu.
Eng: A public subnet has an IGW route, while a private subnet does not.

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5. Can a Subnet belong to multiple VPCs?
No, a subnet can belong to only one VPC.
When we create a subnet, we select the VPC where the subnet will exist.
Simple ga gurthupettuko:
Oka subnet only one VPC ki matrame belong avutundi.
Eng: A subnet can belong to only one VPC.

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6. Can a Subnet span multiple Availability Zones?
No, a subnet can exist in only one Availability Zone.
For high availability, we create separate subnets in different Availability Zones.
Simple ga gurthupettuko:
Oka subnet only one AZ lo untundi, multiple AZs kosam separate subnets create chestham.
Eng: A subnet can exist in only one Availability Zone.

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7. What is a Subnet CIDR?
Subnet CIDR defines the IP address range available for that subnet.
The subnet CIDR must be a smaller range inside the VPC CIDR.
Simple ga gurthupettuko:
Subnet CIDR ante subnet lo use cheyyagalige IP address range.
Eng: Subnet CIDR defines the IP address range of a subnet.

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8. What is the relationship between VPC CIDR and Subnet CIDR?
The VPC CIDR defines the complete IP range of the VPC.
Subnet CIDRs divide that VPC IP range into smaller network ranges.
Simple ga gurthupettuko:
VPC CIDR big IP range, Subnet CIDR aa range lo smaller IP range.
Eng: VPC CIDR is the big IP range, while subnet CIDR is a smaller range inside it.

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9. Can two Subnets have the same CIDR?
No, subnets inside the same VPC cannot have overlapping CIDR ranges.
Each subnet should have its own unique IP range.

Simple ga gurthupettuko:
Same VPC lo subnet CIDR ranges overlap avvakudadhu.
Eng: Subnet CIDR ranges cannot overlap inside the same VPC.

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10. What is a Route Table association with a Subnet?
A subnet is associated with a route table to decide where its network traffic should go.
Different subnets can use different route tables.
Simple ga gurthupettuko:
Subnet ki Route Table associate chesthe traffic ekkadiki vellalo decide avutundi.
Eng: A route table decides where the subnet's traffic should go.

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11. Can multiple Subnets use the same Route Table?
Yes, multiple subnets can be associated with the same route table.
This is useful when multiple subnets need the same routing rules.
Simple ga gurthupettuko:
Multiple subnets ki same Route Table ni associate cheyyachu.
Eng: Multiple subnets can use the same route table.

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12. Can one Subnet use multiple Route Tables?
No, a subnet can be associated with only one route table at a time.
If we change the association, the subnet starts using the new route table.
Simple ga gurthupettuko:
Oka subnet ki same time lo one Route Table matrame untundi.
Eng: A subnet can use only one route table at a time.

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13. What is a Default Route in a Subnet?
A default route sends traffic that does not match a more specific route to a particular target.
For example, 0.0.0.0/0 is commonly used for internet-bound IPv4 traffic.
Simple ga gurthupettuko:
Default Route ante specific route lenappudu traffic vellalsina route.
Eng: A default route is used when no more specific route matches the traffic.

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14. What is a Private Subnet used for?
Private subnets are commonly used for application servers and databases that should not be directly reachable from the internet.
They can still communicate with other resources and access the internet through a NAT Gateway when required.
Simple ga gurthupettuko:
Private Subnet ni direct Internet access avasaram leni servers and databases kosam use chestham.
Eng: Private subnets are used for resources that should not be directly reachable from the internet.

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15. What is a Public Subnet used for?
Public subnets are commonly used for resources that need direct internet connectivity.
Examples include internet-facing load balancers and some public EC2 instances.
Simple ga gurthupettuko:
Public Subnet ni Internet-facing resources kosam use chestham.
Eng: Public subnets are used for resources that need direct internet connectivity.

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16. How do you make a Subnet public?
We associate the subnet with a route table that has a route to an Internet Gateway.
Resources inside the subnet also need the required public IP and security settings for internet access.
Simple ga gurthupettuko:
Subnet ni public cheyyali ante Route Table lo Internet Gateway ki route undali.
Eng: A subnet becomes public when its route table has a route to an Internet Gateway.

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17. How do you make a Subnet private?
We do not give the subnet a direct route to an Internet Gateway.
If internet access is required, we route outgoing traffic through a NAT Gateway.
Simple ga gurthupettuko:
Subnet private ga undali ante direct Internet Gateway route ivvakudadhu.
Eng: A subnet is private when it has no direct route to an Internet Gateway.

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18. What are Reserved IP Addresses in a Subnet?

AWS reserves five IP addresses in every subnet for networking purposes.
These addresses cannot be assigned to resources such as EC2 instances.
Simple ga gurthupettuko:
Prathi subnet lo AWS five IP addresses ni networking kosam reserve chestundi.
Eng: AWS reserves five IP addresses in every subnet.

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19. Why do we create Subnets in multiple Availability Zones?
We create subnets in multiple Availability Zones to improve application availability.
If one Availability Zone has a problem, resources in another AZ can continue running.
Simple ga gurthupettuko:
Multiple AZs lo subnets create chesthe application ki better high availability vastundi.
Eng: Multiple AZ subnets help improve application availability.

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20. What is the most important thing to remember about Subnets?
A subnet is a smaller network inside a VPC and exists in one Availability Zone.
Public or private behavior mainly depends on its routing, especially whether it has a direct route to an Internet Gateway.
Simple ga gurthupettuko:
Subnet ante VPC lopala small network, adi one AZ lo untundi and routing batti public or private avutundi.
Eng: A subnet is a small network inside a VPC, and its routing determines whether it is public or private.